

LMAP — Liquid Media Access Protocol

A Peer-to-Peer Protocol for Media Distribution and Ownership

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Abstract

A purely peer-to-peer system for film distribution would allow digital media ownership to exist independently of any single platform, studio, or streaming service. The current model of centralized streaming creates fundamental misalignments: platforms bear infrastructure costs proportional to viewership, creators receive opaque compensation, and consumers own nothing permanent. We propose a layered protocol in which each film exists as fungible tokens within a unified registry contract, with separate provision for copyright ownership. Token quantity determines usage rights — from personal viewing to theatrical exhibition — at thresholds configured per title. Encrypted content is stored on a content-addressed network. Access to decryption material is gated by current on-chain ownership, evaluated against live chain state by a native access layer, so that access moves with the token. At maturity the access layer is decentralized and no single party can release a key to a wallet that does not hold it; during bootstrap a single foundation issuer performs this function, gating transfer and first-bind — never the playback of content a holder already possesses. For content whose licensing contractually requires endpoint protection, a compliant tier adds hardware-attested per-device key wrapping; a legacy deterministic-key construction survives only for public-domain demonstration content. A network of user-operated nodes — Seeds — distributes content, inverting the conventional cost model as viewers become node operators: above a modest operator attach rate the network strengthens as viewership grows (the precondition is stated plainly in §9, not assumed). Smart contracts automate royalty distribution on every transaction, including secondary sales where collectors can sell exhibition rights directly. The protocol is governed by an independent foundation, structurally distinct from any commercial operator. We argue that liquid media — physical-grade ownership in liquid digital form — is achievable without compromising the sovereignty that makes the design worthwhile.

1. Introduction

The transition to streaming was marketed as disruption, but the underlying economics reveal a fundamentally broken model. When a film succeeds on a centralized platform, infrastructure costs increase proportionally — the platform is economically punished for having popular content. This creates incentives misaligned with the goal of connecting audiences with compelling stories.

The consumer experience has equally degraded. Digital purchases are platform-locked, creating fragmented collections scattered across services. When platforms sunset or lose licensing rights, purchased content disappears. Unlike physical media, there is no true ownership — only temporary access privileges revocable at the platform's discretion.

Creators face similar opacity. Residual payments flow through accounting systems designed to obscure rather than illuminate. The shift from theatrical and physical media sales to streaming has compressed compensation without providing comparable transparency or participation in secondary markets.

What is needed is a protocol-level solution addressing all constituencies simultaneously — analogous to how DVD or USB function: not as a platform a consumer signs up for, but as a *standard* that many implementations interoperate through. We propose LMAP (the Liquid Media Access Protocol): a protocol where digital media is tokenized into standardized units granting verifiable ownership and access rights; content is distributed through a peer-to-peer network that scales positively with demand; smart contracts ensure automatic royalty distribution on every transaction; and ownership is structurally separable from any particular platform that runs on the protocol.

What consumers experience is *liquid media* — a generic public term for the category LMAP enables. Liquid media is to streaming services what physical media was to broadcast television: a different form of ownership, with different rights, that buyers can hold, transfer, and pass on.

2. Token Architecture

Each film exists as a unique token ID within a master ERC-1155¹ registry contract. Unlike ERC-721² non-fungible tokens, where each token is distinct, ERC-1155 allows millions of identical tokens per film, enabling both mass distribution and liquidity. A single contract scales to thousands of films, each identified by a unique token ID, with configurable quantities per title.

The token's URI points to a content identifier on a distributed storage network containing immutable metadata: title, synopsis, credits, and references to media assets. This architecture ensures that even if any particular implementation of the protocol ceases operations, token holders retain permanent access through the decentralized network and the open protocol specification.

Implementation status: shipped on Polygon mainnet as the `WyllohRegistryProtocolV4_1` contract. Verified source available via public block explorers.

3. Separation of Distribution and Copyright

A critical distinction exists between distribution rights and underlying copyright. The ERC-1155 registry manages distribution tokens — fungible units representing viewing and exhibition rights. A separate ERC-721 contract is specified to establish copyright ownership: a single non-fungible token per film representing the intellectual property itself.

The copyright token grants control over derivative works, merchandising, and the ability to propose governance actions such as token splits. Distribution token holders own rights to view, exhibit, and trade — but not to create sequels or merchandise. Crucially, the copyright holder cannot unilaterally mint additional distribution tokens; supply modifications require token holder approval, protecting holders from dilution. This mirrors traditional film finance, where investors hold revenue participation without controlling the underlying intellectual property, while adding protections absent in conventional structures.

This separation enables powerful scenarios. A studio could acquire the copyright token to produce a sequel while existing distribution tokens continue trading. The original filmmaker benefits from the sale; distribution token holders retain their viewing rights and market participation. Rights flow to natural owners without forcing all-or-nothing transactions.

Implementation status: design intent. The current V4.1 registry is the ERC-1155 distribution layer; the ERC-721 copyright registry is specified for a future protocol version (V6) and has not yet been deployed. The current protocol's role-gated minting (see §15) serves as a near-term substitute for the copyright registry's supply-modification controls.

4. Modular Rights Through Stacking

Token quantity determines usage rights through a threshold-based licensing system. A single token grants personal viewing rights — analogous to owning physical media. As tokens accumulate in a wallet, additional rights unlock at thresholds the filmmaker configures at tokenization time.

Thresholds are intentionally not fixed by the protocol; each filmmaker sets thresholds appropriate to their content’s economics. A documentary might set low thresholds to encourage screenings; a genre release might set high thresholds reflecting commercial value. The protocol does not impose a universal price for theatrical rights because no such universal price exists in reality. The market discovers appropriate pricing through trading at the thresholds the filmmaker has chosen.

For illustrative purposes only — not as protocol-level constants — typical configurations might include:

- *1 token* — personal viewing rights
- *100 tokens* — small-venue exhibition
- *60,000 tokens* — theatrical exhibition rights with access to professional distribution formats

But these are choices, not specifications. Thresholds for any given film are encoded in that film’s metadata at minting and are immutable thereafter for the life of that token’s contract.

This mechanism creates natural price discovery for commercial rights. Rather than negotiating complex licensing agreements, distributors acquire sufficient tokens on the open market. Token supply is fixed at initial minting; if market conditions warrant increased liquidity, token holders may vote to approve a split, multiplying token quantities while preserving each holder’s *proportional ownership*. The copyright holder may propose splits but cannot execute them unilaterally. A split does **not** leave the rights tiers untouched, and this is by design: because thresholds are fixed absolute token counts (§4), a 2× split halves every tier’s cost as a *fraction of supply*, uniformly — so the **ratios between tiers are preserved** (theatrical stays 60,000× a personal token) while access to the whole ladder gets cheaper together. That expanded accessibility is the point of a split, not a side effect. One honest consequence to name: since a holder may bind up to *balance* copies (§7.2), a split proportionally raises how many simultaneous copies each holder can run — a split is itself a form of authentic-supply expansion, which is precisely how the protocol answers scarcity-driven piracy (§14). A useful frame for this deliberate accessibility increase is a **wide theatrical release**: lowering the bar to entry widens distribution of the same work — the split trades per-token price for reach, by the holders’ own vote. New creative works — sequels, adaptations, alternative cuts — are tokenized as separate token IDs with independent supplies, so the original token’s market stays *distinct*. One honest limit: per-ID supply is protected, but nothing prevents the same film being re-minted as a “remaster” under a *new* ID (under V5, permissionlessly, by anyone). On-chain provenance disambiguates the authentic edition from a copy, but a competing edition’s market impact can still land; the protection is *distinctness and provenance*, not economic exclusivity against every re-issue.

Implementation status: shipped. Stacking thresholds and on-chain enforcement are operational in V4.1.

5. Layered Architecture

The protocol decomposes into eight layers. Each layer has a defined responsibility, a published interface to the layers above and below it, and an implementation status. Different parties can build different layers as long as the interfaces are honored. Conflating layers — particularly the entitlement layer with the cryptographic layer — is how comparable projects have failed.

#	Layer	Responsibility
0	Trust Anchors	Cryptographic primitives; hardware roots; wallet identity; contract immutability
1	Storage	Content distribution substrate (IPFS, manifests)
2	Cryptography	Encryption, key hierarchy, threshold-mediated key release, dispersal, watermarking
3	Entitlement	Smart contracts: distribution + copyright registries, royalties, rights tiers
4	Network	Peer discovery, shard request/response, LAN streaming
5	Attestation	Compliant-tier hardware attestation (the open tier gates access at Layer 2)
6	Application	Storefronts, library UI, playback clients, integrator tooling
7	Governance	Foundation structure; federated certification; protocol governance

A storefront (Layer 6) only needs to read entitlement (Layer 3); a Seed manufacturer needs to implement Layers 1, 2, 4, and optionally 5; an academic auditor of the compliant tier only needs to engage with Layer 5 and the cryptographic claims of Layer 2. Each layer should be specifiable, testable, and replaceable independently of the others. The protocol does not mandate that any single party implement all eight layers.

The remaining sections of this paper traverse these layers from storage upward.

6. Content Storage

Media files are encrypted using AES-256-GCM before distribution on a content-addressed storage network³. The metadata structure nests content identifiers: the token URI points to a metadata file containing references to both unencrypted promotional assets (poster, trailer) and encrypted media files. Each encrypted asset specifies a threshold — the token quantity required for access — corresponding to the rights tiers established at minting.

Encrypted content is stored in a chunked format (`[Length(4 bytes)][IV(12 bytes)][Ciphertext+Tag(16 bytes)]` per 4-MiB chunk) which is self-authenticating per chunk and supports streaming decryption with constant memory. The format is documented in the protocol specification; any conformant client can implement encryption and decryption against it.

A future enhancement at this layer is *threshold dispersal of ciphertext shards* — distributing encrypted content shards across the Seed network using Reed-Solomon erasure coding, with the option of secret-share-based reconstruction⁴, such that reconstruction requires a quorum of nodes. This is a resilience property at the storage layer, not the load-bearing security mechanism (see §7.4 for the precise scope of this claim).

Implementation status: chunked AES-256-GCM encryption and IPFS storage are shipped. Threshold ciphertext dispersal is specified for a future protocol version.

7. Cryptography and the Two-Tier Access Model

The protocol gates access in two ways, for two kinds of content. For permissive and indie content (the *open tier*), access is gated cryptographically at Layer 2 by threshold-mediated key release — honest access control that requires no special hardware on the client side. For content whose licensing contractually requires endpoint protection (the *compliant tier*), the protocol adds hardware-attested per-device key wrapping at Layer 5. Both tiers run the same protocol underneath; they differ only in what stands between a holder and the plaintext.

A note on one word, used precisely throughout: **“commercial” means *sold for money*** — as opposed to free or public-domain content. It is *not* a synonym for either tier. Commercial content spans both: paid indie content sold on the open tier, and premium or studio-licensed content whose contracts require endpoint protection on the compliant tier. When this paper means the latter specifically, it says *premium* or *endpoint-protection-required*, not “commercial.”

7.1 Open Tier

Any conformant Seed implementation serves content licensed under permissive terms — public domain, Creative Commons, indie creators who opt their content into open-tier distribution.

The legacy deterministic key-derivation construction. Earlier versions of this whitepaper described the open tier’s content key as derivable from public on-chain data using the formula `SHA-256(contract_address || ":" || token_id || ":wylloh-v1")`, with the encrypted master key delivered to a buyer by a storage service after that service verified on-chain ownership. This construction was articulated as a deliberate design choice in service of platform independence. Reviewed against the requirements of any content of commercial value, it has been retired as a production model: the derivable-key construction does not actually gate decryption by ownership, because any party with the public chain data can derive the wrapping key. The construction is appropriate for *demonstrating the protocol’s other mechanics on public-domain content* — the role it played in the V4.1 deployment of *The Cocoanuts* (1929, public domain) — and survives only for that purpose. New commercial content does not tokenize under this construction.

Production access control: threshold-mediated key release. LMAP’s production access-control mechanism is *threshold-mediated key release*. A master key is generated as 32 random bytes at content preparation, used to AES-256-GCM-encrypt the film in the chunked wire format documented in §6, then encrypted to a Distributed Key Generation public key held by a threshold cryptographic network. The threshold-encrypted master key is stored alongside the film’s IPFS metadata. The film’s metadata declares an Access Control Condition — typically `balanceOf(:userAddress, :tokenId) > 0` against the LMAP registry contract.

At decryption, the client signs a SIWE-style authentication message proving control of the requesting wallet. The client submits the authentication, the threshold-encrypted master key, and the Access Control Condition to the threshold network. Threshold-network nodes independently verify the wallet signature, query the chain for current Access Control Condition state, and release decryption shares only if the condition evaluates true. The client collects *T* shares and reassembles the master key, then decrypts the film using the standard chunked AES-GCM mechanism.

Transfer behavior emerges from the architecture. When a token moves from one wallet to another, the Access Control Condition is re-evaluated against current chain state at the next decryption attempt. The seller’s wallet no longer satisfies `balanceOf > 0`; the buyer’s now does. No re-wrapping, no foundation involvement, no seller cooperation required.

The threshold-release primitive is substrate-independent: any threshold-decryption network that can evaluate a Polygon Access Control Condition and return decryption shares can serve it. LMAP’s production access layer is **native** — the same federated framework that issues compliant-tier attestation credentials (§7.3) operates and progressively decentralizes the threshold network, so no single external service stands between a holder and content they own. Where a mature external threshold network is used as an interim substrate during bring-up, it is exactly

that: a swappable bridge behind a stable interface, never a dependency the protocol's guarantees rest on. The protocol's rule is that the access layer must not be rentable from, or revocable by, any party outside the federated framework.

What the open tier provides honestly. Threshold-mediated key release is *access control*: it gates which wallets can obtain a decryption key. It cannot prevent a legitimate holder, once they hold the key and the bytes, from saving plaintext to disk, capturing the screen, recording the HDMI output, or otherwise creating an unauthorized copy of content they legitimately decrypted. Defense against that case rests on social incentive, the ease of legitimate purchase, the alignment of buyers with creators, forensic watermarking (§7.5), and — where licensing contracts demand it — the compliant tier (§7.2). The open tier is approximately DVD-level protection at the playback edge: cryptographic enforcement of access; honest acknowledgment that the analog hole and post-decryption copy are not prevented.

A resale consequence, stated plainly. Because permanence caches the master key locally after first decryption (§8), an open-tier holder who has *ever* played a film keeps a working copy and key even after selling the token. The open tier therefore has **no copy scarcity**: what an open-tier secondary buyer acquires is the license, the on-chain provenance, and future access — explicitly *not* exclusivity against a seller who has already decrypted. This is the deliberate open-tier posture, and the same one physical media took; enforceable one-copy-per-token scarcity is precisely what the compliant tier's binding (§7.2) adds, which is why premium content that requires it uses that tier.

7.2 Compliant Tier

Studios and rights holders who require contractual digital-rights management cannot use the open tier as-is. The compliant tier provides hardware-attested per-device key wrapping for content where this is required.

In the compliant tier:

- Each compliant Seed holds a per-device keypair generated inside its hardware secure element. The private key is non-extractable.
- Content keys are wrapped to each compliant Seed's public key. Compromising one compliant Seed yields *that Seed's local content only* — a constant per-compromise bound with no catalog-scale master secret. (Precisely: per-compromise yield does not grow with network size; per-title leak probability does grow with holder count, bounded by watermark attribution and revocation, not cryptography — §7.4.)
- Wrapped keys are unwrapped only inside the secure element. The unwrapped key never appears in main memory or on persistent storage outside the secure perimeter.
- Issuance of a wrapped key requires the Seed to present a fresh *attestation report* signed by its secure element, containing current firmware measurements. The report is verified against the Seed's attestation credential before any wrapped key is issued.

Reference hardware for the compliant tier includes ARM TrustZone⁵-capable SoCs with discrete secure elements (e.g., NXP SE050 or equivalent), or platforms with TPM 2.0⁶ support. Remote attestation⁷ follows established patterns from contemporary trusted-computing literature. The key hierarchy is precise: each title has an **independently random** master key — there is no catalog-wide master (that would be the class break §7.4 relies on not existing). Within a title, HKDF with domain separation⁸ derives per-asset and per-rendition subkeys, so keys are never reused across assets or titles. Where a title carries multiple **rights tiers** (§4), each tier's asset is keyed and protected *independently*, under its own access condition (`balanceOf(:userAddress, :tokenId) ≥ threshold_T`), such that obtaining a lower tier's key never derives a higher tier's; the exact construction is fixed in the forthcoming technical specification.

Hardware binding is a normative requirement of the compliant tier. A compliant-tier implementation **MUST** use a hardware secure element to (a) generate the per-device keypair, (b) hold the private key in non-extractable form, (c) perform key unwrapping, and (d) sign attestation reports. An implementation that performs any of these

operations outside a hardware secure element is not conformant with the LMAP compliant-tier specification and MUST NOT be marketed as such. This is the protocol-level guarantee studios rely on when authorizing content for the compliant tier. The wire-level specifics that make this checkable — the attestation report format, the attestation credential schema, the issuer federation protocol, and a machine-checkable conformance suite — are **in active specification as launch deliverables** (see Implementation Status), not yet published; until they are, “conformant” is a stated requirement, not something a third party can yet build and verify against.

A clarifying distinction: only the *decryption* step requires compliant hardware. Encrypted compliant-tier content is freely distributable through the same content-addressed substrate as open-tier content — any device can hold, pin, or serve it. Tokens remain freely transferable across all wallets. The compliant tier gates *playback*, not *carriage*. This preserves user sovereignty (a buyer’s ownership is unaffected by hardware availability; they can transfer their token at any time) while satisfying the studio requirement (decryption is impossible without an attested secure element). The architectural model mirrors theatrical distribution under DCI: the encrypted DCP is portable and freely distributable; only certified projectors with KDM-bound keys can play.

One active copy: binding, release, and free exchange. Per-device wrapping gives the compliant tier a property the open tier does not have — enforceable scarcity, furnished as a *tool* rather than imposed as a lock. A wrapped key is *bound* to one secure element at a time. Because the entitling token is a *fungible* ERC-1155 unit, there is no distinguishable “the token” to flag; binding is tracked instead as a **count per (wallet, tokenId)** — `boundCount` — under the invariant `boundCount ≤ balance`, so a wallet’s transferable units are `balance - boundCount`. This count lives in an **on-chain binding registry from launch**: a separate, world-readable accounting contract, so any platform, wallet, or prospective buyer can verify a unit’s released status without trusting a private ledger. **Only the player that bound a copy can release it — without revealing which device it is.** A **bind** is recorded only when it carries a valid, non-revoked compliant-device attestation *and* the owner’s wallet signature, and the registry enforces `boundCount ≤ balance` by reading the token’s on-chain balance. The write stores a **blinded binding commitment** — the hash of a fresh secret the device holds — *not* the device’s identity. A **release** is then authorized by *opening that commitment* (revealing the secret): opening proves the writer is the very device that bound, so it needs no wallet signature and exposes no device identity. This is what lets the **device self-audit release a copy the wallet has sold** — the departed wallet need not co-sign, and no observer learns which player held it. No central issuer writes the flag; the attested fleet does, from launch, and the contract enforces the invariant. The one wallet-authoritative write is **recovery**: after the report window, the owner force-releases their *own* (`wallet, tokenId`) units directly — the registry is indexed by wallet, so the owner enumerates their own titles and a lost device need never appear. A single party *can* delay a release by revoking a device’s write capability, pushing that holder onto the recovery path; it *cannot* prevent one, precisely because recovery is wallet-authoritative. The issuer’s remaining role is confined to *key wrapping* — enabling decryption at first bind — separate from flag state and decentralizing on its own path (§7.3). The token remains a plain ERC-1155 throughout, and the registry is a distinct contract that never touches transfers, so tokens minted at launch carry forward unchanged.

The registry informs; it never gates. The entitling token transfers freely on-chain at all times — a gift, an inheritance, or a direct peer sale needs no one’s permission — and the protocol does not let the issuer, or any party, block the exchange of an owned token. That is the boundary the protocol will not cross. Safe trade is then built on the tool by choice: to sell without residual a holder *releases* a unit (the bound device deletes its wrapped key and attests it; `boundCount` decrements); an escrow contract can settle a sale atomically against the registry, so a buyer transacting through it never pays for a still-bound unit; and a platform may list only released units, or surface a `bound` warning on the rest. These are platform policies resting on a protocol-furnished flag — where the protocol sets down and platforms pick up (§15). The protocol supplies the tools for coordination and honest disclosure; it lets the market decide where to use them.

A bound copy plays indefinitely offline; the binding is recorded, never re-verified, so playback never contacts the network. A cooperating, online device also *self-audits* at its own next transaction — shedding any bound title whose wallet no longer owns it — so an actively-used device cannot retain copies it has sold; this check runs only at voluntary transactions and on confirmed non-ownership, never during playback. If a bound device is lost, sold, or destroyed, playback of its already-bound content is unaffected (it fails open) and the token’s transferability is recovered by an owner-signed **report** that releases the titles after a 30-day window — **no heartbeat or presence beacon**; the device verifiably erases the released titles as a precondition of its next transaction, and a refusing device is revoked. This yields one-copy-per-token scarcity *for trade conducted through honest platforms and escrow*, with no continuous ownership check — enforcement only at the discrete moments of binding and release. Two residuals remain, harmless to a buyer’s guarantee of a working title: a compromised device that reports `released` while retaining its copy; and — because raw transfer always works — a still-bound unit moved outside an escrowed sale, which the registry surfaces as `boundCount > balance` for that wallet. Both are physical-media-tier personal leakage rather than a redistribution vector, and their *bound* — “forensically watermarked, hence attributable” — is **conditional** on integrating a watermark scheme with measured robustness (§7.5), not a delivered property. They are the cost of preserving free exchange, priced in rather than designed out.

7.3 Attestation Issuers and the Federated Framework

Attestation credentials in the compliant tier are signed by independent *attestation issuers*. An issuer is a party that verifies a Seed’s hardware integrity and signs a credential binding the device’s public key to a manifest of approved firmware measurements. Issuers in turn operate under a federated *attestation framework* governed by the Liquid Media Foundation (see §15).

No single issuer can unilaterally approve or deny attestation; the framework’s governance can revoke any issuer whose practices fail audit. Studios negotiate with the framework, not with a single platform operator. The model is structurally analogous to the federation of certificate authorities that underpins the public web’s TLS infrastructure: independent issuers, audited practices, root trust managed by a body distinct from any commercial participant.

The framework launches with a single issuer (the Liquid Media Foundation itself) and a public commitment to federate as the network matures. Honesty requires naming the gap: the specific trigger — a node count, device count, date, or maturity metric — is **not yet set**, and until it is published this is a commitment to federate, not a *defined* milestone. Fixing that number, together with the first-audit and admin-key-sunset milestones (§15), is itself a near-term deliverable; the intent is a visible, measurable trigger rather than an indefinite promise.

7.4 What Each Tier Provides — Access Control vs. Endpoint Protection

A precise statement of what each tier achieves, kept honest because the two tiers solve different problems and should not be confused on the same axis.

Open tier (threshold-mediated key release). This is *access control*. Threshold release correctly enforces that a wallet can obtain a decryption key only while that wallet currently holds the relevant token. Transfers cause access to move with the token; the seller’s next decryption attempt fails the on-chain check; the buyer’s succeeds. The protocol’s rule is that no single party — not the foundation, not Wylloh, not the protocol’s own contracts — may release a key to a wallet that does not currently satisfy the access condition. At maturity this rule is enforced directly by a distributed substrate of cryptographic nodes native to the protocol, with no single party able to break it; during bootstrap a single Foundation issuer performs the release under that same rule, a temporary centralization the decentralization roadmap (§7.3) dissolves. In neither case does issuer or network liveness gate the *playback* of content a holder already possesses (§8) — only first release and transfer. **What threshold release does not do:** it does not, and no software-only mechanism can, prevent a legitimate holder from capturing or copying the plaintext stream once they have legitimately decrypted it. That is the analog hole and the post-decryption copy. Threshold release does not address those threats — and we do not claim that it does.

Compliant tier (attested per-device key wrapping — the binding model). This adds *endpoint protection* on top of ownership-gated access, and is the mechanism the flagship launch is built on (§7.2). Both are provided together by binding: the issuer verifies current ownership and the device’s attestation, then wraps the master key to the device’s secure element. Crucially, the compliant player is a **sealed direct player** that renders and outputs video itself (§9) — the decrypted stream is confined to a protected media path to the player’s *own* HDCP-protected output and is never streamed in the clear to a separate device. Key custody and attestation live in a hardware secure element; the bulk media decrypt runs in a protected media pipeline (a TEE) inside the same sealed perimeter, so the raw key and decoded frames never leave that perimeter. This constrains what the device does with the stream *against the device’s own owner* — what licensing contracts for premium content require.

On parity with commercial DRM — stated carefully. Hardware- attested endpoint protection is the category that also includes Widevine L1, PlayReady, and FairPlay. LMAP’s compliant tier belongs to that category by construction, but the protocol does **not** claim measured robustness *parity* with those specific systems, which rest on SoC-integrated secure-video paths, protected memory, and certified provisioning refined over many years. Parity is a property to be *earned* through a published robustness specification and independent review (a forthcoming compliant-tier deliverable, §7.2), not asserted. Until then the honest claim is narrower and still substantial: a sealed direct player with a hardware secure element, a protected media path, and attested output — materially stronger than software-only access control, and scoped to the independent and boutique premium content the launch serves.

These are not equivalent [access vs. endpoint]. They solve different threat models: threshold release gates *who gets the key*; the compliant tier’s sealed player additionally gates *what the device does with the key*. They are not substitutes. Threshold release at the open tier is honest and sufficient for indie, public-domain, Creative Commons, and creator-opt-in content, where extraction at the playback edge is acceptable defense resting on social incentive plus watermarking. For content whose licensing demands endpoint protection, the compliant tier — a sealed direct player — provides it.

A modest supplementary claim — threshold dispersal of ciphertext (storage layer). Sharding ciphertext across Seeds raises the per-shard extraction cost. An attacker compromising one Seed obtains a fraction of the ciphertext for any given film; reconstruction requires reaching the quorum threshold. This is real but bounded: a sufficiently motivated attacker can target enough Seeds to reach quorum. Threshold dispersal provides resilience and modestly raises extraction cost; it is not a load-bearing security claim.

7.5 Watermarking

Forensic watermarking binds a leak to a specific wallet, raising the cost of unauthorized distribution beyond the bare cryptographic extraction cost. Where it is used, the insertion locus differs by tier, and robustness is not yet established — so claims elsewhere in this paper that lean on watermark attribution (“bounded,” “traceable,” “attributable”) are **conditional** on integrating a scheme with independently measured robustness, not asserted as already delivered.

Compliant tier. Watermarking inserts in the **protected media path** — the TEE media pipeline (§7.4), at decode time, before any frame reaches an output — *not* in the discrete secure element, which holds keys and can neither bus nor decode video. Each playback binds the holder’s identity to the frames within the sealed perimeter. Transcoding-robust marking requires commercial watermarking IP (e.g., Verimatrix, NAGRA, Irdeto); until such a scheme is integrated and independently measured, the attribution-based residual bounds of §7.2 remain conditional.

Open tier. The open tier does **not** rest its defense on watermarking. Per-user marking collides with content addressing — per-wallet bytes mean per-wallet CIDs, which breaks the byte-identical-CID guarantee and the shared-swarm distribution model and reintroduces a per-download central serving path; client-side insertion is in any case unenforceable in a tier that welcomes arbitrary third-party clients. The open tier instead accepts playback-edge

permeability by design (§7.4) and relies on social incentive, easy legitimate purchase, and creator alignment. A content-addressing-compatible open-tier mark (e.g., segment-level side-channel marks) is an open research item, not a v1 dependency.

The protocol specifies watermark *insertion points* but mandates no particular scheme; commercial implementations with established studio relationships exist, and robustness against transcoding and adversarial filtering remains an ongoing research area.

Implementation status: chunked AES-256-GCM encryption and IPFS storage are shipped on Polygon mainnet via the V4.1 registry. *The Cocoanuts* (1929, public domain) was tokenized and decrypted end-to-end across two independent architectures (macOS arm64, Raspberry Pi aarch64) using the legacy deterministic key-derivation construction. The **compliant tier** — attested per-device binding on the sealed direct player (§7.2) — is the launch access-control mechanism for the first, premium, endpoint-protection-required release, consistent with a Seed-gated first title. (Paid indie content that needs no endpoint protection is *also* commercial, and rides the open tier once its threshold-release migration lands — see the Implementation Status.) Its wire-level attestation specification, conformance suite, and the Foundation’s first (single) issuer are the highest-priority near-term deliverables of the launch build; hardware binding is declared normative (§7.2). **Native threshold-mediated key release** serves the open (no-hardware) tier and is in active migration off the retired deterministic construction in the Wyllow reference implementation; the native, fully-decentralized threshold network — and the decentralization of the compliant-tier issuer from single → federated → threshold — is the longer-horizon endgame (§7.3).

8. Permanence Guarantees

The protocol commits to a permanence guarantee in both tiers: a Seed that has legitimately acquired a license to a film at any point in time retains the ability to play that film indefinitely, regardless of subsequent network availability, attestation revocation, or platform fate.

In the open tier, this guarantee is engineered through three mechanisms operating in combination. First, after a client has successfully decrypted a film once via threshold-mediated key release, the master key is cached locally — in the browser’s persistent storage, the Seed daemon’s encrypted local storage, or any conformant client implementation’s local persistence. Subsequent playback of the same film does not contact the threshold network; the client decrypts directly from local state. Second, holders may export their master keys as user-encrypted backups — wrapped with a passphrase the holder chooses, written to any storage the holder controls. A backup file plus the downloaded encrypted bytes plus the user’s passphrase enables playback in perpetuity, even if all foundation and threshold-network infrastructure ceases to exist. Third, the ownership record on the chain remains the durable source of truth; a holder who has lost both local cache and backup can re-acquire access through any conformant implementation of the protocol that still operates against the chain, because the wallet’s holdings are recoverable from public data.

For the legacy V4.1 deployment of public-domain content under the deterministic key-derivation construction (§7.1), permanence is even more direct: the decryption key for any owned film is derivable from public on-chain data using the documented formula. A Seed unplugged for twenty years and plugged back in can play any *Cocoanuts*-era token it previously legitimately acquired, with zero network connectivity, using a reference client a third party authored from the public protocol specification. This property is the one the original deterministic construction was articulated to preserve; we keep it for the public-domain tokens that depend on it, while threshold release supersedes the construction for new content.

The canonical claim, stated precisely: *downloaded films play forever offline; only re-download, first-bind, and transfer require the access layer to be live — at maturity with no single-company point of failure, and during bootstrap gated by a single foundation issuer whose liveness affects transfer and acquisition, never playback of what a holder already possesses.*

In the compliant tier, this guarantee is engineered. Attestation gates the *issuance* of wrapped keys, not the *playback* of content already keyed. A Seed whose attestation credential is later revoked retains the wrapped keys it previously received and continues to play the corresponding films. Revocation prevents the Seed from acquiring further content through the compliant tier; it does not retroactively disable the Seed’s existing library. This is the architectural answer to the rugpull pattern that has characterized streaming services and digital media stores.

The hardware itself remains a single point of failure: a Seed whose secure element fails physically loses access to its locally-wrapped keys. The protocol mitigates this by treating the on-chain ownership record as the durable source of truth. A user whose Seed has died can re-acquire wrapped keys for their owned films onto a new compliant Seed, because the wallet’s holdings are recoverable from the blockchain. **The blockchain remembers; the hardware is replaceable.** Physical media has no equivalent recovery path when a disc fails, so at maturity this is a stronger guarantee. Stated with its conditions, since it is a comparison: recovery requires a compliant issuer to be alive and willing to re-wrap (at bootstrap, one Foundation-operated issuer), a compliant player still being manufactured, and — for a lost or dead player — the 30-day recovery window (§7.2). Physical media’s disc failure has no recovery, but also no counterparty; the compliant tier trades a counterparty dependency for a recovery path, and is honest that the trade is only fully in the holder’s favor once the issuer is decentralized (§7.3).

9. The Seed Network

Content distribution operates through *Seeds* — user-operated nodes that combine encrypted-content storage, peer-to-peer participation, and playback. Seeds run on commodity hardware following published reference designs.

A Seed is a **sealed direct player**: it connects to the television and renders and outputs video itself, so that compliant-tier content is decrypted and displayed entirely within one sealed device — the only place endpoint protection (§7.4) is coherent. Decryption happens inside the player’s secure element and protected media path, and plaintext never crosses the network. The reference design is a fanless, small-form-factor player with a custom case and CEC so the television’s own remote drives it. Earlier designs explored a *headless* Seed that served decrypted video to thin clients (Roku, Apple TV, iOS) on the home network; that model reaches devices users already own but forfeits endpoint protection at the client hop, so it is retained only as a **secondary mode for open-tier content** — never the path for compliant-tier premium content, which plays on the sealed player alone. (Streaming to a personal device from one’s *own* Seed is a convenience feature layered on the same footing: available for open-tier content, out of scope for the compliant tier’s guarantees.)

Each Seed stores content its operator owns, plus optionally contributes overflow capacity to the network. Popular films accumulate across more Seeds as more users own tokens; niche content maintains availability proportional to its audience. No central coordination determines what Seeds store; ownership patterns drive distribution.

This architecture inverts conventional streaming economics — with a dependency we state plainly. Centralized platforms bear infrastructure costs proportional to viewership: success is expensive. In the Seed network, viewers who *operate a Seed* add capacity as they add demand, so per-stream cost falls as the operator fleet grows. The inversion is therefore a property of an *operator* network, not of browser distribution: browser-only buyers add demand without adding capacity, and the cost curve inverts only above a modest **operator attach rate** — the fraction of viewers who run a Seed rather than watch through a hosted client. We do not yet publish a measured figure for that rate or for residential-uplink performance at 4K HDR bitrates; the claim is directional, and its precondition is named rather than assumed. Two things make the attach rate realistic: the reference Seed’s standalone utility (§16) gives viewers a reason to own one independent of any network reward, and the Foundation treasury funds baseline pinning redundancy (§10.1) so availability never depends on attach rate for low-operator or long-tail titles. Seeds participate by default because their primary value to the operator is local — the films they own,

accessible to LMAP-compatible playback clients on their local network — so network participation is a beneficial side effect rather than the purchase rationale. This is precisely the property that distinguishes a Seed from the token-incentivized storage networks whose hardware had no standalone value (§16).

A reference Seed device — the *Origin* — is specified separately from this paper as a buildable open-hardware design. The protocol does not require the Origin or any particular Seed implementation; any conformant device can participate.

Implementation status: an open-source reference Seed daemon is operational. It verifies wallet ownership directly against the on-chain registry, fetches encrypted content from IPFS (preferring a co-resident Kubo node where present, falling back to public gateways), decrypts streaming with constant memory, automatically pins the source CID to make the Seed a provider for the content it holds, and exposes a long-running HTTP service with mDNS auto-discovery and HTTP byte-range streaming for LAN client applications. End-to-end operation has been demonstrated on independent ARM64 hardware (Raspberry Pi 4), with byte-identical decrypted output across architectures and with no dependency on any centralized service. The reference *Origin* device is in pre-production. The web client at wylloh.com serves as the first reference application.

10. Marketplace Mechanics

The protocol includes integrated marketplace functionality. Primary sales occur through minting at prices set by the title's author. Secondary sales enable peer-to-peer trading with automatic royalty distribution on each transaction.

10.1 Three-Way Revenue Split

Every primary sale routed through the protocol distributes payment across three parties, with proportions set per title at mint time:

- **Protocol fee: 2.5%** (immutable, hardcoded at the contract level). Flows to the Liquid Media Foundation treasury for shared infrastructure: IPFS pinning redundancy, contract auditing, security work, open-source maintenance.
- **Publisher fee: 0–25%** (set per title, immutable for that title). Flows to the publisher — the wallet that minted the title. Publishers are entities that bring works to market: established storefronts, indie collectives, or self-publishing authors. Their cut compensates curation, marketing, audience development, and brand value.
- **Author share: remainder.** Flows to the author — the credited creator(s). Distributed via an optional royalty shareholders configuration (§10.3) supporting up to 50 collaborators per title.

For a typical configuration — 10% publisher fee — the split is 2.5% / 10% / 87.5%. Authors receive dramatically better economics than traditional publishers (where author shares of 10–20% are common). The protocol's 25% publisher cap is the only opinionated stance: *publishing on LMAP shall not become as extractive as the systems it replaces.*

10.2 Publisher and Author as Distinct Roles

The protocol records two distinct identities per title:

- **Publisher** is set automatically from the wallet that calls the mint function. Cryptographically authentic — cannot be impersonated. Establishes the brand identity associated with the title and receives the publisher fee on primary sales.
- **Author** is provided at mint and identifies the credited creator of the work. Receives the author share of every primary and secondary sale.

The two roles are often the same wallet (an author who self-publishes) but can differ (a publisher mints under their brand and routes the author share to a separate creator wallet). This mirrors the publishing-house / author distinction familiar from books, films, and music.

10.3 Royalty Shareholders

Authors can configure up to 50 royalty shareholders per title, each receiving a configurable percentage of the author's share. This supports the realistic case where a tokenized work has multiple contributors deserving compensation: a film with director, lead actors, composer, screenwriter; a video game with developers, artists, and key team members; an album with songwriter, performers, and producer.

Shareholder allocations are **append-friendly but not revocable**: the author can *add* collaborators, or increase a share by drawing from the author's own residual remainder, but **cannot reduce or zero an allocation already granted** to another wallet without that wallet's consent. This keeps the flexibility to bring collaborators on as the work evolves while making a granted residual a durable commitment — a contributor's stream, once given, cannot be unilaterally taken back. That is consistent with the immutability guarantee (§15) and is the on-chain-residuals promise a contributor can actually rely on: cast and crew are paid on every sale, forever, and no later edit can quietly cut them out. Sum of shares can be $\leq 100\%$ of the author's share; any remainder flows to the author wallet, preserving the author as the residual recipient. For projects with more than 50 beneficiaries, shareholders compose with external splitter contracts (e.g., 0xSplits): one shareholder slot can route to a splitter address that distributes further.

Distribution happens automatically on every protocol-routed sale — both primary and secondary. No manual royalty disbursement, no escrow, no off-chain ledger reconciliation.

10.4 Royalty Enforcement Boundary

The protocol enforces royalties cryptographically for sales routed through its marketplace functions. External marketplaces honoring the ERC-2981 royalty standard will respect the configured royalty when displaying and processing sales. Direct wallet-to-wallet transfers (the gift, inheritance, and personal-give use cases) do not trigger royalties — this is a property of the underlying ERC-1155 standard and a deliberate preservation of token-as-property semantics.

Off-chain payment with on-chain gift is theoretically possible and unenforceable cryptographically, as with every royalty-bearing token system. The protocol's mitigation is structural: the low 2.5% protocol fee minimizes the savings from circumvention; the integrated marketplace makes legitimate sales easier than coordination; on-chain transparency makes wash-trade patterns visible and reputationally costly.

10.5 Bulk Acquisition for Rights Stacking

The marketplace supports atomic batch purchases (`batchBuyListings`), enabling a single transaction to acquire tokens from many sellers simultaneously. This is essential for the rights-stacking use case (§11): a buyer assembling tokens to meet a theatrical-exhibition threshold may need to acquire thousands of tokens from hundreds of sellers in a single coordinated purchase. The atomic semantics guarantee all-or-nothing: if any listing in the batch fails, the entire transaction reverts, preventing the partial-fill case where a buyer pays fees but doesn't reach their target threshold.

Platform UIs aggregate, sort, and present available listings; the protocol provides the batch-execution primitive.

10.6 The Protocol Fee Is Not a Platform Fee

The 2.5% protocol fee supports shared infrastructure all implementations depend on, identical across every LMAP-compatible marketplace. Third-party marketplaces may layer their own commercial markup on top, competing on user experience, curation, and discovery rather than by undercutting the protocol fee. This places third-party

marketplaces and the reference web client at wylloh.com in a peer relationship with one another, not a competitive one.

10.7 Settlement and Gas Abstraction

Transactions settle in stablecoin to insulate buyers from cryptocurrency volatility. The current implementation uses USDC.e (bridged USDC) on Polygon. Settlement on Polygon means that buyers must currently hold both USDC.e and a small amount of POL to cover network gas — friction the protocol acknowledges as a near-term limitation rather than an intended feature. The architectural answer is account abstraction via ERC-4337 *paymaster contracts*: a paymaster pays POL gas on the user's behalf, recovering the cost via a small USDC surcharge bundled into the transaction. The reference contract V5 exposes paymaster hooks allowing a paymaster contract to be wired in by foundation governance. Paymasters are deployed and funded at the marketplace layer; each marketplace can choose whether and how to subsidize gas as part of its commercial offering.

Implementation status: primary sales with three-way distribution, secondary sales with royalty enforcement, royalty shareholder distribution, batch acquisition, and the protocol fee are specified in the V5 reference contract. The V5 contract is in active development; deployment under foundation governance is a 2026 milestone. The legacy V4.1 registry remains in operation for existing tokens (open-tier only, two-way author/protocol split) during the transition.

11. Commercial Rights Discovery

The stacking mechanism enables a novel form of commercial rights acquisition. Consider a film that achieves cult status: tokens are distributed across thousands of fans who purchased during initial release. A theater chain wishes to conduct a theatrical run, requiring tokens above the exhibition threshold the filmmaker configured for that title.

Rather than negotiating with the original filmmaker, the theater acquires tokens directly from fans through open market purchases. As buy pressure increases, token prices rise, rewarding early believers who funded the film's success. Royalties trigger on each transfer, flowing to the copyright holder regardless of who sells. The market determines that theatrical rights to this particular film, at this particular moment, command a specific price — no negotiation required.

Commercial exhibition requires tokens to be staked for the exhibition period, and it enforces that stake through a **distinct, more-stringent regime from personal viewing — on the same fungible token**. A personal copy binds and plays offline forever, its playback never checking ownership (§7.2, §8) — correct, because the residual it leaves is one personal copy. Exhibition cannot use that fails-open model: a bound-and-abandoned *exhibition* copy would retain a whole commercial rights package, not a personal one. So exhibition playback is **licensed, not sovereign** — a venue is a **licensee**, and its entitlement is enforced at screening time. The *exact mechanism is deliberately left open*: DCP-grade exhibition content is far more sensitive than home content, and the exhibition interface must be **developed directly with the DCI community** to meet the industry's established requirements — engaging DCI/KDM frameworks rather than reinventing them — not asserted unilaterally here. What is firm is the **tiered unlock model** and the two-regime split, which scopes the personal tier's "playback never checks" rule to personal copies rather than contradicting it. One token, two unlocks; and the liquidity flows between them — a collector's tokens can aggregate into a venue's exhibition stake, giving a sleeper film its theatrical run *after* it has already circulated digitally, then disperse back to collectors.

Implementation status: design intent, **forthcoming** — the launch focus is home viewing. Stacking thresholds are operational at the contract level today; the exhibition regime — its enforcement mechanism, its DCI/KDM interface, DCP-grade asset protection, and the staking/tier accounting — is **TBD, to be worked directly with the DCI community** (the tiered unlock model and the two-regime split are settled; the mechanism is not — see the device compliance note §2.1).

12. Production Funding

The token architecture enables presale-based production financing. Rather than relying on studio executives or platform algorithms to greenlight projects, creators can presell tokens directly to audiences. This validates market demand before production begins while distributing ownership of the work directly to its audience.

Presale tokens grant the same ownership rights as post-release tokens but are purchased at a discount, rewarding early supporters. Smart contracts can structure milestone-based fund releases with multi-signature approval from stakeholder representatives. If production fails to meet milestones, remaining funds return to presale participants.

This model removes centralized gatekeepers from greenlighting decisions. A film need not fit a platform's content strategy. If sufficient audience interest exists to fund production, the project proceeds. The presale serves simultaneously as market research, financing mechanism, and community-building exercise.

Implementation status: design intent. The token architecture supports presales architecturally; the milestone-based escrow contracts and the corresponding presale-tokenization workflow are specified for a future protocol version.

13. Privacy

Contemporary streaming platforms extract extensive personal data to drive recommendation algorithms and advertising. The protocol operates on a different principle: *movies should not watch their viewers*. Analytics derive exclusively from public blockchain data — transaction patterns, token distribution, market activity — rather than viewing behavior or personal information.

The conventional streaming model links user identity to viewing history to platform to advertisers, forming a chain of surveillance maintained primarily for monetization through inferred consumer preferences. The protocol model maintains pseudonymous wallets executing public transactions on a transparent ledger, with **no telemetry from playback events and no viewing patterns aggregated by any party** — *movies never watch their viewers*, the claim the protocol enforces absolutely: playback is a local operation that contacts nothing (§8), in the compliant tier as much as the open one (a bound player never emits a presence signal, so recovery is report-triggered, not surveillance-detected).

What the protocol does **not** promise is anonymity of *ownership*, and it is honest about the difference. Ownership and transfers are public by design; so is the binding registry (§7.2), which records **bind and release events per (wallet, tokenId) with timestamps** — a public account of which wallets acquired copies of which titles and when, and of the transferability lifecycle. The registry is deliberately built to expose *no more* than that: binding commitments are **blinded**, so **device identity never appears on-chain** and the wallets sharing a household player are not linked through it; and nothing in it records *viewing*. Pseudonymity is also not unlinkability — a wallet funded through a KYC'd fiat ramp, or correlated by chain analysis, can be tied to a real person by a determined party; the protocol's guarantee is the absence of a *viewing-surveillance* chain, not the impossibility of deanonymizing a public ledger. The issuer learns *acquisition* events — which on-chain ownership already largely reveals — but never *when* or *how often* a holder watches; oblivious issuance, in which the issuer does not learn the title, is a specified future privacy enhancement. Creators receive aggregate insights — token distribution by region inferred from public data, secondary-market activity, holder concentration — without compromising individual *viewing* privacy.

14. Security Philosophy

The protocol does not attempt perfect copy protection. History demonstrates that technological restrictions on digital media inevitably fail while degrading legitimate user experience. Instead, the system aligns incentives such that ownership provides value beyond mere access — liquidity, permanence, community membership — making legitimate acquisition the rational choice.

The analog hole — capture of content via camera or screen recording during legitimate playback — remains possible. Watermarking is the only mitigation, and watermark robustness against transcoding is an ongoing arms race. The protocol does not claim immunity from this class of attack; it claims that legitimate purchase confers economic benefits that bare ciphertext extraction does not.

When content is owned not by a distant corporation but by potentially many individual token holders, piracy dynamics shift. Unauthorized distribution harms not an abstract studio but a community of fellow enthusiasts. Token holders own copies of the work itself, aligning their interest with its integrity rather than its piracy. Social contracts, properly constructed, provide stronger protection than technological locks against the casual case; threshold-mediated key release provides honest cryptographic access control against the non-holder case; and the compliant tier provides hardware-attested endpoint protection against the case where licensing contracts require it. These mechanisms address different threat models, and the whitepaper is precise about the distinction: threshold release gates *who gets the key*; hardware attestation in the compliant tier gates *what the device does with the key during playback*. The two are complementary in the compliant tier; threshold release alone is sufficient and honest for the open tier.

Physical media operated on similar principles. Copy protection on optical discs was trivially circumvented, yet the market thrived for decades. Convenience, quality assurance, and social norms around ownership provided sufficient protection. The protocol adds financial incentives physical media lacked: ownership appreciates in genuinely successful work, and secondary markets reward early adoption.

A specific guidance for technical readers: when evaluating the protocol's claims, separate three distinct claims that should not be conflated. (1) The *access-control claim*: threshold-mediated key release correctly gates decryption by current on-chain ownership. At maturity it is evaluated by a distributed network of cryptographic nodes with no single party able to release a key to a non-holder; during bootstrap a single Foundation issuer performs the release under that same rule (§7.3), its liveness affecting transfer and first release, never playback of content already held. (2) The *endpoint-protection claim*: hardware-attested per-device key wrapping at the compliant tier constrains what the playback device can do with the decrypted stream, addressing the legitimate-viewer-attempting-capture threat that no software-only mechanism can address. (3) The *operational claim*: legitimate purchase is more economically rational than extraction for participants aligned with creators. Each claim holds at its own layer under its own threat model. The whitepaper is precise about this because the audiences this document serves — investors, developers, studio business affairs — read the same words and need to extract the same meaning. The protocol does not claim that threshold-mediated release is equivalent to or stronger than hardware-attested DRM; the two solve different problems. The protocol claims that threshold release is honest access control, that hardware attestation is the right capability where endpoint protection is contractually required, and that the combination of ownership, transferability, permanence, and platform independence delivered to collectors is a categorically different — and, for this audience, better — set of properties than streaming or account-locked stores provide.

15. Governance and the Liquid Media Foundation

The protocol is governed by an independent foundation, structurally distinct from any commercial operator. The foundation holds the protocol's intellectual property under a permissive open-source license, stewards the compliant-tier attestation framework, publishes the protocol specification, and operates as a protocol-neutral standards body.

Commercial entities that build on the protocol — storefronts, hardware manufacturers, marketplace operators — are participants in the LMAP ecosystem rather than its operators. This structural separation mirrors precedents in adjacent ecosystems: the Filecoin protocol governed by the Filecoin Foundation, structurally distinct from Protocol Labs; the Ethereum protocol governed independently from any single Ethereum company. The pattern preserves credible-neutrality: filmmakers, third-party storefronts, hardware manufacturers, and eventually studios can build on the protocol without fearing that the protocol's stewards are also their commercial competitors.

The foundation, at its mature governance, will steward:

- **Specification governance.** Protocol upgrades, EIP-equivalent proposals, and version transitions.
- **The attestation framework.** Maintaining the federation of attestation issuers, audit standards, and revocation governance for the compliant tier (§7.3).
- **Treasury.** Custody of the protocol fee revenue, application to shared infrastructure costs, and grants to ecosystem development.
- **Token-holder governance.** Supply-split proposals (copyright-holder-proposed, holder-approved), which multiply token quantities while preserving proportional ownership. A title's thresholds, fees, and royalty terms are immutable for the life of the token, and a royalty share already granted to a collaborator is irrevocable (§10.3 — additions are allowed, reductions are not); new terms mean a new mint, not an amendment, and distribution rights are never withdrawable by vote (§8).

A pragmatic note about the present: the foundation does not yet exist as a formal legal entity. The current governance posture is that of a stewarded bootstrap, in which the founding team operates the protocol on the implicit promise of formalizing the foundation and transitioning to community governance over a multi-year arc. This is a commitment, not a fact, and the protocol's roadmap includes the foundation's formal establishment, the attestation framework's first audit milestones, and the public sunset of any admin keys held by the founding team.

Curation: gateless, not absent. The protocol layer is permissionless — any wallet can mint a title (V5 onward). Curation happens at the application and registry layers, not at the protocol. Three signals work together:

1. **Cryptographic identity at mint.** The wallet that mints a title is recorded as the publisher, on-chain, immutably. Brand identity is wallet-bound and cannot be impersonated.
2. **Foundation-maintained verified publishers registry.** The Liquid Media Foundation publishes a curated list of verified publishers at `lmap.org/publishers.json`. Verification signals quality and accountability without restricting the protocol; unverified publishers can still mint, but won't appear in verified-only application views.
3. **Application-layer editorial choice.** Each platform built on LMAP applies its own filter: the Wylloh storefront shows titles from Wylloh and other respected publishers; an indie aggregator might show all unverified content; a music-focused platform filters by `mediaType: music` and music-publisher wallets. The protocol exposes the truth (who minted what); applications shape the experience.

This model — *permissionless protocol, curated applications* — mirrors how DNS works: anyone can register a domain, but resolvers, browsers, and search engines each apply their own filters. Quality emerges economically rather than by gatekeeping. A publisher with poor taste mints work nobody buys; a publisher with good taste accumulates verified status and earns the publisher fee from real demand.

The legacy V4.1 registry retains its `FILM_CREATOR_ROLE` gate as the editorial mechanism for the open-tier-only deployment serving already-minted titles. New tokenizations under V5 land in the permissionless model.

Implementation status: the foundation is in formation (incorporation as a 501(c)(6) trade association is a funded milestone of the upcoming raise); V5's permissionless minting is specified and implemented in the reference contract scaffold; deployment under foundation governance is a 2026 milestone; the verified publishers registry will launch concurrently with V5 deployment.

16. Comparison with Prior Work

LMAP inherits ideas from several adjacent ecosystems and addresses limitations specific to each.

Centralized streaming services (Netflix, Disney+, et al.) own the relationship with audiences and bear infrastructure costs proportional to viewership. They cannot offer durable digital ownership because durable ownership conflicts with subscription economics. LMAP inverts this by routing ownership through a public ledger

and distribution through a peer-to-peer network.

Kaleidescape ships hardware-backed ownership for theatrical-quality content, but operates within a closed ecosystem at luxury price points and serves a narrow demographic. LMAP shares the philosophical commitment to genuine ownership but extends the addressable audience by separating the protocol from any particular hardware vendor and accepting standard 4K HDR streaming quality rather than requiring lossless masters.

Plex demonstrates local, home-owned media at scale (over 25 million users), but with primarily user-ripped content of unclear provenance, no rights or royalty flow, no curatorial focus, and a closed-source business model. The Seed shares Plex's core virtue — your library lives at home, not in a rented cloud — while differing in form (a sealed direct player, not a headless server feeding third-party clients) and in substance: on-chain provenance, automatic royalties, and an open protocol anyone can implement.

Filecoin¹⁰ and **Helium** demonstrate decentralized-network plays with token-incentive mechanisms. Both attracted speculative participants more than utility-driven users, primarily because the hardware in each case had no standalone consumer value. The purchase rationale collapsed if the token economics did not deliver expected returns. The reference Seed device — the Origin, built by Wylloh as the first commercial implementation of LMAP — has a primary consumer utility: the operator's film library, accessible to LMAP-compatible playback clients on their local network. This utility does not depend on any token-incentive mechanism. Network participation is a beneficial side effect of a device that operators would buy regardless. This sidesteps the chicken-and-egg adoption problem that has limited prior decentralized-CDN plays.

Bitcoin¹¹ and **Ethereum**¹² establish the cryptographic and economic primitives this protocol relies upon. The protocol's contributions specific to film distribution are the dual-contract architecture (§3), the modular rights-stacking mechanism (§4), the two-tier access model (§7), the permanence guarantee (§8), and the Seed-network distribution model (§9).

17. Conclusion

We have proposed a protocol for film distribution addressing fundamental misalignments in centralized streaming. By tokenizing films into standardized units with embedded licensing rights, we create genuine digital ownership. By distributing content through a peer-to-peer network that grows stronger with demand as viewers become operators (§9), we invert economics that punish platforms for success. By automating royalty distribution including on secondary sales, we ensure transparent creator compensation through markets that never existed for physical media.

The two-tier access model resolves the perceived conflict between open-protocol architecture and contractual studio trust requirements. The open tier gates access cryptographically through threshold-mediated key release — honest access control that moves with the token and depends on no single party — while accepting that endpoint extraction is not prevented, the same posture physical media took for decades. The compliant tier provides hardware-attested per-device key wrapping for content where contractual DRM is required, governed by a federated attestation framework structurally independent from any commercial operator.

The separation of distribution tokens from copyright ownership enables fluid rights markets while preserving creator control over intellectual property. Commercial exhibitors can acquire rights directly from collectors. Studios can acquire copyright for derivative works without disrupting existing token markets. Rights flow to natural owners through market mechanisms rather than negotiated deals.

The protocol requires no permission from existing gatekeepers. Public-domain content bootstraps the network, demonstrating capabilities before commercial titles join. As the network grows, established platforms face a choice: implement the protocol and benefit from its efficiencies, or cede ground to peer marketplaces and competitors that do.

LMAP represents not disruption but recoherence — using peer-to-peer technology once deployed to undermine the industry to instead restore fair compensation and sustainable economics. The same distributed networks that enabled unauthorized copying can now enable legitimate, liquid, permanent ownership. *In the venom, the antidote.*

Implementation Status Summary

A precise summary of the protocol's components by their status as of this paper's publication:

- **Shipped (legacy V4.1 deployment, scoped to public-domain content):** ERC-1155 distribution registry on Polygon mainnet at `0x8e834c6031A2d59e1cADd24Ab24F668301ed7c1D`; role-gated minting; modular rights stacking; chunked AES-256-GCM encryption; legacy deterministic key-derivation construction (§7.1) — retired as a production model and now scoped to public-domain demonstration content only; self-contained manifest reading from on-chain metadata + IPFS, with storage-service fallback; open-source reference Seed daemon (verify ownership, fetch + decrypt, auto-pin, LAN-stream over HTTP/mDNS); web client at `wylloh.com`; first tokenized film (*The Cocoanuts*, 1929, public domain) sold and downloaded end-to-end across two independent architectures (macOS arm64, Raspberry Pi aarch64) with byte-identical output. The V4.1 contract continues to serve existing tokens under the legacy construction.
- **In active migration (blocking open-tier commercial-content onboarding):** migration of the Wylloh reference web client and Seed daemon from the legacy derivable-key construction to native threshold-mediated key release. Completion of this migration is a prerequisite for tokenizing commercial content *on the open tier* (paid indie content needing no endpoint protection); it does not gate the premium, compliant-tier first release, whose launch mechanism is attested binding (below). The V4.1 deployment continues to serve already-tokenized public-domain content under the legacy construction during and after the migration. The Wylloh reference implementation scopes the open-tier framing described in §7.1 as a deliberate brand position rather than as a tolerated limit — DRM-free post-decryption is disclosed as a field in standard listing terms and marketed as vinyl-grade ownership, with the social and economic incentives of legitimate acquisition treated as load-bearing product logic. That framing is implementation-level, not protocol-level; other compliant implementations may scope the open tier differently.
- **Specified, in active implementation (V5):** the LMAPRegistryV5 reference contract for deployment on Polygon mainnet — adds the three-way revenue split (§10.1), publisher/author distinction (§10.2), royalty shareholder distribution (§10.3), atomic batch acquisition for rights stacking (§10.5), `mediaType` field for media-agnostic operation, updateable platform treasury via multi-sig timelock, ERC-4337 paymaster hooks, and permissionless minting (§15). Compatibility with threshold-mediated key release is the production access-control mechanism. Scaffold with passing tests; full marketplace implementation, comprehensive test coverage, and deployment under foundation governance are 2026 milestones. **Not yet in the V5 feature set** (it predates the binding model): the **binding registry** contract and the **binding-aware escrow/settlement** path (§7.2) — the compliant tier's buyer guarantee reads against these; they are launch-critical interfaces still to be specified, tracked with the attestation flow below rather than assumed complete in V5.
- **Specified, in implementation (hardware, clients, and the compliant tier):** the reference Seed device (the *Origin*), a sealed direct player; its native client applications for open-tier LAN streaming (Roku, Apple TV, iOS); and the **compliant-tier launch mechanism** — attested per-device binding (§7.2) with a single Foundation issuer. The wire-level attestation specification (report format, credential schema, revocation flow), the conformance suite, and the binding and marketplace-settlement flow are launch deliverables in active specification.
- **Specified, future protocol versions (V6+):** ERC-721 copyright registry; on-chain staking for commercial-exhibition windows; presale-funding milestone-escrow contracts; threshold dispersal of ciphertext shards; **federation of the attestation issuer and decentralization of the compliant-tier key service from single** →

federated → **threshold** (§7.3). The on-chain binding registry (§7.2) is itself a launch component, written by the attested player fleet (no central flag-writer); the protocol adds no transfer-veto — exchange stays free by design.

- **Long-horizon (operational maturity required):** studio-grade compliant-tier engagement with major rights holders.

This paper is a working specification. Components shipped today behave as described; components specified are commitments the protocol is engineered toward. The Liquid Media Foundation, when formally established, will maintain this status summary as a living public artifact.

Changelog

Version 2.6 (July 2026). Completes the access-control rework begun in the v2.5 draft, following two rounds of external review. v2.5 was a pre-publication draft; this entry records the delta rather than rewriting v2.5's, keeping the correction record honest. Substantive changes: (1) **The compliant-tier binding model is finalized.** Binding state moved from an *off-chain issuer-held bound/released flag* to an **on-chain, world-readable binding registry present from launch**, tracking a **boundCount per (wallet, tokenId)** (not a per-token flag) written by the **attested device fleet** under a **blinded-commitment rule** — a bind records `hash(secret)` and never device identity; a release is authorized by *opening* that commitment, so a device can release its own sold copy with no wallet co-signature and no identity exposure. The **reverting-transfer-hook endgame is retired**: the registry *informs, it never gates*, the token transfers freely on-chain always, and no party — issuer included — can block exchange (a single party can at most *delay* a release by revoking a device, never withhold it). A **device self-audit** sheds sold titles at the device's next voluntary transaction (never during playback). (2) **Exhibition split into a distinct regime.** Personal copies fail open (sovereign); exhibition playback is *licensed, not sovereign*, with the enforcement mechanism deliberately left TBD to be developed with the DCI community — resolving the personal-bright-line-vs-§11 collision. (3) **Privacy (§13) re-derived** from the new architecture (public bind/release ledger, blinded so device identity stays off-chain; “no link to real-world identity” softened for KYC ramps and chain analysis). (4) **Fee economics** stated as *97.5% of the list price, rights-holder-side* with the variable publisher fee set by the minting platform. (5) **Governance/immutability/shareholders** reconciled: supply-split votes only, distribution rights never withdrawable, granted royalty shares irrevocable (additions allowed), splits honestly re-price rights tiers while preserving proportional ownership. (6) **README and companion documents aligned**; version/status/legal-structure/watermark-locus/carriage corrected throughout. Wire-level attestation, the binding registry and binding-aware escrow, and the exhibition mechanism are named as launch deliverables in active specification, not asserted as built.

Version 2.5 (July 2026). Forward-ports the access-control refinements settled after v2.4. (1) **The access layer is native.** The threshold-release primitive is stated as substrate-independent (§7.1); no specific external threshold network is named as the production substrate. The protocol's rule is that the access layer must not be rentable from, or revocable by, any party outside the Foundation's federated framework; an external network may serve only as a swappable interim bridge behind a stable interface. (2) **The “certified tier” is renamed the “compliant tier”** throughout, naming the tier by the open conformance standard a device meets rather than by an authority's grant. (3) **A binding model is introduced for the compliant tier (§7.2)** — per-device wrapping tracks a `boundCount` per `(wallet, tokenId)` so a copy binds to one secure element at a time, plays offline indefinitely, and is *released* before a unit trades without residual, restoring one-copy-per-token scarcity without any continuous ownership check. *(The v2.5 draft carried an earlier form of this model, including an on-chain bound/released status; the mechanism — a fleet-written blinded-commitment registry that informs but never gates, with the transfer-hook endgame retired — was finalized in v2.6, above.)* Companion documents (`PROTOCOL_LAYERS.md`, `PROTOCOL_POSITIONING.md`) are being brought into alignment (a multi-pass rework in progress).

Version 2.4 (June 2026). Supersedes v3.0 (also June 2026) following external architectural review. v3.0 attempted to add an LMA protocol token, a dedicated Liquid Media Chain (Polygon CDK zero-knowledge rollup), token-incentivized storage and threshold networks, an offshore Cayman Foundation Company structured to issue the token, and a three-year dependency-sunset roadmap. External review identified this scope as disproportionate to the protocol’s actual use case: the protocol’s value does not require a token economy; Lit Protocol already provides the threshold-release primitive without LMAP needing to build its own; the Filecoin/Helium chicken-and-egg risks v3.0’s §17 itself cited as cautionary would have applied to LMAP’s own token; and the resulting securities exposure was premature overhead for a token the architecture did not require. v3.0 also contained a category error: claims that threshold cryptography provides security “equivalent to or stronger than hardware-attested DRM.” That claim conflated access control (which threshold release provides — gating who gets the key) with endpoint protection (which hardware DRM provides — what the device does with the decrypted stream during playback, against the device’s own owner). The two solve different threat models and are not directly comparable. v3.0 is archived as superseded ([whitepaper/archive/whitepaper-v3.0-superseded.{md,pdf}](https://whitepaper.archive.whitepaper-v3.0-superseded.{md,pdf})).

v2.4 returns to the v2.3 baseline and forward-ports the substantive cryptographic improvements that survive the review: (1) **§7.1 Open Tier rewritten** — the legacy deterministic key-derivation construction is retired as a production model and scoped to public-domain demonstration content (*The Cocoanuts*); threshold-mediated key release via Lit Protocol’s Naga mainnet becomes the production access-control mechanism. (2) **§7.4 rewritten** as “What Each Tier Provides — Access Control vs. Endpoint Protection”; the prior “Security That Scales with Network Size” framing is retired in favor of an explicit statement that threshold release and hardware DRM solve different threat models. (3) **§8 Permanence Guarantees expanded** with the local key cache, self-wrapped backup export, and on-chain ownership recovery mechanisms; the canonical claim “*downloaded films play forever offline; re-download and transfer require the access network (Lit) to be live; no single-company point of failure*” is now stated explicitly. (4) **§14 Security Philosophy sharpened** to distinguish the access-control claim, the endpoint-protection claim, and the operational claim — three distinct claims that should not be conflated. **The Implementation Status Summary** is updated to reflect the Lit migration as the blocking near-term engineering milestone and the legacy derivable-key construction as retired-to-public-domain-scope. **The §16 comparison with prior work** is unchanged; we explicitly do not build a Filecoin/Helium- shaped token-incentive layer atop the same cautionary analysis.

A consistency pass harmonized the framing sections that the v2.3 baseline had carried forward unchanged — the abstract, the §5 layered-architecture table, the §7 section opener and title (now “the Two-Tier Access Model”), the §9 Seed-incentive description, and the §17 conclusion — with the threshold-release production model described in §7.1 and §7.4. Those sections previously described the retired derivable-key open tier as the protocol’s production mechanism and omitted threshold-mediated key release. Stale gas-token naming (MATIC → POL) was corrected in §10.7.

The Wylloh Seed product line referenced in this document evolved during May 2026 implementation work into two reference SKUs (Seed One and Origin Seed); the conceptual reference Seed described in §9 is unchanged in protocol terms, but readers should be aware that the implementation now ships in two form factors. The protocol/implementation distinction (LMAP as Apache-2.0 spec, Wylloh as the first implementation) is unchanged.

Version 2.3 (May 2026). Reference Seed daemon promoted from “specified” to “shipped” in the Implementation Status. §9 (Seed Network) updated to reflect end-to-end operation of the open-source reference daemon on independent ARM64 hardware (Raspberry Pi 4) with byte-identical decrypted output across architectures and no dependency on any centralized service. The daemon’s operating behavior — IPFS fetch with local Kubo preference, automatic content pinning to make the Seed a provider, LAN-streaming via HTTP/mDNS with byte-range support — is documented as the canonical reference behavior for compliant Seeds. §7.2 (Compliant Tier) gains a normative statement: hardware binding (SE-resident keypair generation, non-extractable private keys, in-SE key unwrapping, in-SE attestation signing) is a MUST for V6 conformance, and an implementation omitting any of these operations is not LMAP-V6-conformant and MUST NOT be marketed as such. A clarifying paragraph distinguishes *playback*

(which requires compliant hardware) from *carriage* (which does not — tokens, encrypted content, and pinning remain freely accessible across all hardware), mirroring DCI theatrical distribution. Wire-level attestation specifics (report format, credential schema, issuer federation protocol, revocation flow) are deferred to a separate LMAP V6 technical specification, forthcoming.

Version 2.2 (May 2026). V5 reference contract specification additions. §10 (Marketplace Mechanics) substantially expanded to specify the V5 economic model: three-way revenue split (protocol 2.5% / publisher 0–25% / author remainder), distinct publisher/author identities (publisher cryptographically authentic via mint-time `msg.sender`), royalty shareholder distribution supporting up to 50 collaborators per title, atomic batch acquisition for rights stacking, and an honest articulation of the royalty enforcement boundary (V5-marketplace sales enforce cryptographically; direct ERC-1155 transfers preserve the gift / inheritance use case). §15 (Governance) updated to reflect V5’s permissionless minting model — curation moves from a protocol-level role gate to a three-signal application-layer model: cryptographic publisher identity, a foundation-maintained verified publishers registry at `lmap.org/publishers.json`, and individual platforms’ editorial choice. Implementation Status restructured to reflect V4.1 (legacy, in operation) and V5 (in active implementation).

Version 2.1 (May 2026). Rebrand pass under Liquid Media Foundation stewardship. The protocol — previously referred to internally as “Wylloh” — is renamed *LMAP* (Liquid Media Access Protocol) to formalize the separation between the open standard (LMAP, stewarded by the foundation) and the first commercial implementation (Wylloh, the company building the reference hardware and storefront). Substantive technical content of v2.0 is preserved unchanged. References to commercial-platform UX details that exceed protocol scope have been generalized. A more substantive v3.0 with deepened §15 (foundation governance) and expanded §16 (comparison with prior work) is forthcoming.

Version 2.0 (April 2026). Comprehensive revision following the April 2026 mainnet launch and the architectural pivot to the headless-Seed network. Key additions: §5 (layered architecture), §7 (two-tier attestation model with attestation issuers and the federated framework), §8 (permanence guarantees), §15 (governance and the foundation), §16 (comparison with prior work). The cryptographic claims of v1’s §6 have been rewritten in v2’s §7 to reflect the actual two-tier model rather than the threshold-key-sharing scheme v1 described aspirationally. The original V1 PDF is preserved in the project archive.

Version 1.0 (April 2026). Initial publication.

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